MAHESWARI PLYWOOD & HARDWARE

Billing and Inventory Management System

Project Analysis Document

Use Case • System Architecture • ER Diagram • Flowchart • UI Wireframes

Prepared as a system-design reference based on the supplied billing application.

1. Project Overview

The Maheswari Plywood & Hardware Billing and Inventory Management System is designed to support the daily operations of a hardware and plywood shop. It brings together sales billing, product and stock management, customer records, supplier purchases, expenses, reports, returns, and shop settings.

The supplied application includes modules for Dashboard, New Sale Ticket, Inventory & Stock Management, Invoice History & Returns, Customer Ledger & Credit (Khata), Purchase & Vendor Management, Daily Expense Tracker, Sales & Profit Analysis, and Shop Settings & Backup Vault.

2. Main Users and Functionalities

User / Actor	Main responsibilities
Shop Owner / Admin	Manage products, stock, customers, suppliers, purchases, expenses, reports, settings, and backups.
Cashier	Create bills, add products, collect payments, hold bills, and print or share invoices.
Customer	Receive invoices, make payments, and review outstanding balances through the shop ledger.
Supplier	Supply products and receive purchase/payment records from the shop.

3. Use Case Diagram

The use case diagram shows how the main users interact with the billing system. The owner/admin maintains the shop data and oversees reports. The cashier handles sales transactions. Customers and suppliers participate in billing, payments, and purchase-related activities.

Use case view (text diagram)

```
MAHESWARI BILLING SYSTEM
  +-------------------------------------------------------+
  |  (Manage Products & Inventory)     (View Reports)     |
  |  (Create Sales Invoice)             (Manage Customers)    |
  |  (Collect Payment)                  (Manage Suppliers)     |
  |  (Process Returns)                  (Record Purchases)    |
  |  (Manage Expenses)                  (Settings & Backup)   |
  +-------------------------------------------------------+
       ^             ^              ^              ^
       |             |              |              |
   Shop Owner      Cashier       Customer       Supplier
   / Admin
```

Actor interactions

• **Shop Owner/Admin:** manages inventory, customers, suppliers, purchases, expenses, reports, settings, and backup.

• **Cashier:** creates invoices, records payments, handles held bills, and processes sales-related tasks.

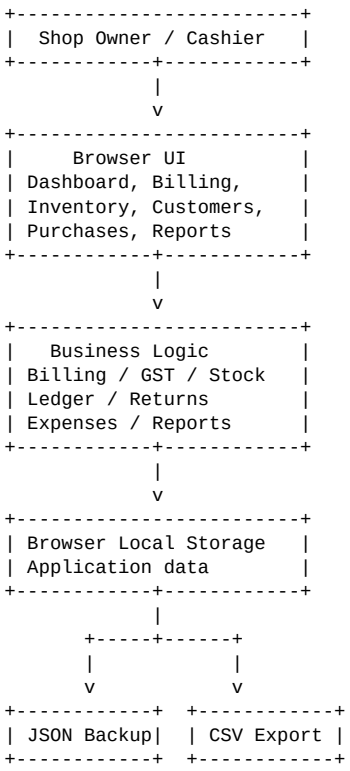
• **Customer:** receives invoices, makes payments, and has transactions recorded in the customer ledger.

• **Supplier:** supplies goods and is associated with purchase and outstanding-balance records.

4. System Architecture

The supplied application works as a browser-based, standalone system. Its user interface and business operations run in the browser, while application records are stored in browser storage. The application supports data backup/export. This means the current design does not require a separate application server for its basic local operation.

Current application architecture

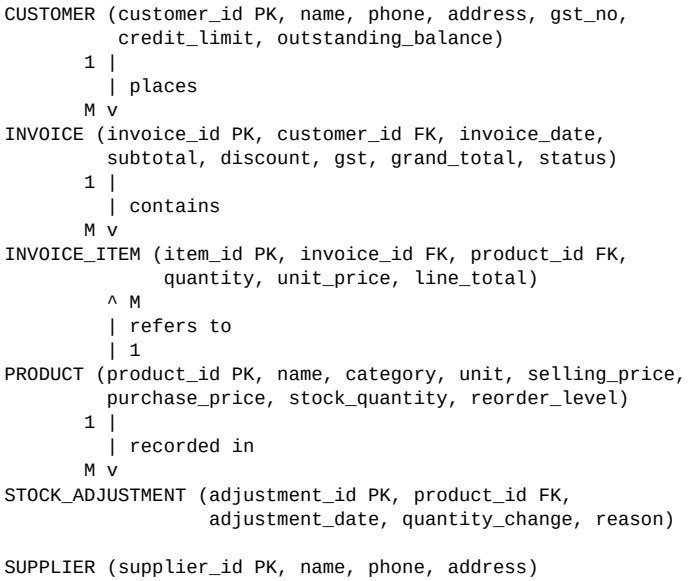


Important architecture note: Browser storage is suitable for a standalone/local-use application. If the shop needs multiple devices to share live data, the system would need a centralized backend and database, for example a Node.js/Express API with PostgreSQL or SQLite. That is a proposed future enhancement, not a feature to assume is already present.

5. Entity Relationship Diagram (ERD)

The following is a logical database design for the system. It represents the main records and relationships needed for billing, inventory, purchases, customer payments, expenses, and stock tracking. The supplied application currently stores its data in browser storage; this ERD is a structured model for documentation or a future database implementation.

Logical ER diagram



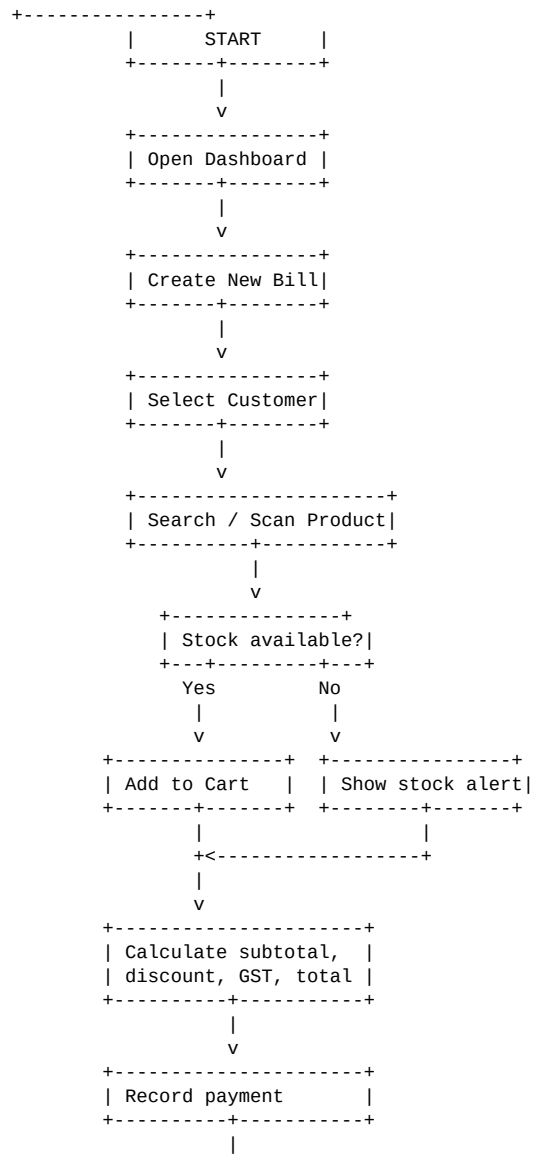
```
       1 |
         | supplies / receives
       M v
PURCHASE (purchase_id PK, supplier_id FK, purchase_date,
          total_amount, paid_amount, balance)
       1 |
         | contains
       M v
PURCHASE_ITEM (purchase_item_id PK, purchase_id FK,
               product_id FK, quantity, unit_cost)

CUSTOMER 1 ---- M PAYMENT (payment_id PK, customer_id FK,
                           invoice_id FK optional,
                           payment_date, amount, method)

EXPENSE (expense_id PK, expense_date, category,
         description, amount)
```

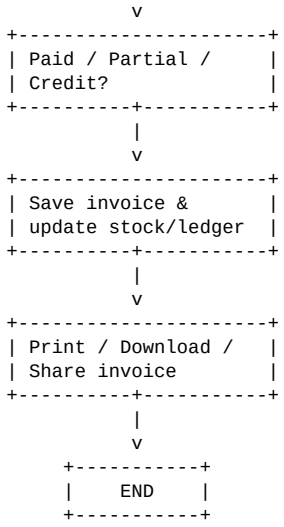
Primary keys uniquely identify records. Foreign keys connect related records. One customer can have many invoices and payments; one invoice can contain many invoice items; one product can appear in many invoice and purchase items; and one supplier can be linked to many purchases.

6. Billing Flowchart

This flow describes the main sales process from opening a new bill to saving and issuing the invoice.

Sales billing process





7. UI Wireframes

The following low-fidelity wireframes show suggested screen layouts based on the modules and functions identified in the supplied application. They are structural sketches, not exact screenshots of the original interface.

7.1 Dashboard

Dashboard layout

```
+-------------------------------------------------------------------+
| MAHESWARI PLYWOOD & HARDWARE                 Search   Profile     |
+----------------------+--------------------------------------------+
| Dashboard            |  TODAY'S SALES   INVOICES   PRODUCTS  LOW  |
| New Sale             |  +-----------+  +--------+ +-------+ +----+ |
| Inventory            |  |  Amount   |  | Count  | | Count | |Qty | |
| Invoice History      |  +-----------+  +--------+ +-------+ +----+ |
| Customers / Khata    |                                            |
| Purchases / Vendors  |  Sales Overview                            |
| Expenses             |  +--------------------------------------+ |
| Reports              |  |          Sales chart                  | |
| Settings / Backup    |  +--------------------------------------+ |
|                      |  Recent Invoices | Low Stock Products      |
+----------------------+--------------------------------------------+
```

7.2 New Sale / Billing Screen

Billing screen layout

```
+-------------------------------------------------------------------+
| NEW SALE TICKET                                  [Hold] [Save]    |
+-------------------------------------------------------------------+
| Customer: [ Search / Select Customer                ] [Add New]   |
| Product:  [ Search product / scan barcode            ] [Add]      |
+-----------------------------------+-------------------------------+
| CART                              | BILL SUMMARY                  |
| Product | Qty | Price | Total     | Subtotal:       ______        |
| Item 1  |  2  | _____ | _____     | Discount:       ______        |
| Item 2  |  1  | _____ | _____     | GST:            ______        |
| Item 3  |  3  | _____ | _____     | Grand Total:    ______        |
|                                   | Payment:        ______        |
|                                   | Balance:        ______        |
+-----------------------------------+-------------------------------+
| [Clear Cart] [Save Invoice] [Print] [Download] [Share]            |
+-------------------------------------------------------------------+
```

7.3 Inventory Screen

Inventory layout

```
+-------------------------------------------------------------------+
| INVENTORY & STOCK MANAGEMENT                [Add Product]         |
+-------------------------------------------------------------------+
```

```
| Search product: [_____] Category: [All v]       |
+-------------------------------------------------------------------+
| Product | Category | Unit | Selling Price | Stock | Status          |
|---------|----------|------|---------------|-------|-----------------|
| Item A  | Plywood  | Sheet|       ____    |  ___  | In stock        |
| Item B  | Hardware | Nos  |       ____    |  ___  | Low stock       |
| Item C  | Boards   | Sheet|       ____    |  ___  | In stock        |
+-------------------------------------------------------------------+
| Selected product: Details / Edit / Delete / Adjust Stock          |
+-------------------------------------------------------------------+
```

7.4 Customer and Ledger Screen

Customer management layout

```
+-------------------------------------------------------------------+
| CUSTOMERS & KHATA                              [Add Customer]     |
+-------------------------------------------------------------------+
| Search customer: [_____]                                          |
+------------------------------+------------------------------------+
| CUSTOMER DIRECTORY           | CUSTOMER LEDGER                    |
| Name | Phone | Balance       | Customer details                   |
| A    | ____  | ____           | Outstanding balance: ______        |
| B    | ____  | ____           | Invoice history                    |
| C    | ____  | ____           | Payment history                    |
|                              | [Record Payment] [Reminder]        |
+------------------------------+------------------------------------+
```

7.5 Reports and Settings

Reports and settings layout

```
+--------------------------------+  +--------------------------------+
| REPORTS                        |  | SHOP SETTINGS                  |
+--------------------------------+  +--------------------------------+
| Date range: [From] [To]        |  | Shop name / address            |
| Sales summary                  |  | Contact / GST details          |
| Profit analysis                |  | Invoice preferences            |
| Expense summary                |  | Tax settings                   |
| Stock report                   |  | Backup / Restore               |
| [Export CSV] [Print Report]    |  | [Save Settings]                |
+--------------------------------+  +--------------------------------+
```

8. Key Modules

Module	Purpose
Dashboard	Provides a summary of sales, invoices, products, and low-stock information.
Billing	Creates sales tickets, calculates totals, records payments, and issues invoices.
Inventory	Maintains product details and stock quantities.
Invoice History & Returns	Displays past invoices and supports sales returns.
Customer Ledger	Stores customer details, credit information, payments, and outstanding balances.
Purchases & Vendors	Records purchases, supplier information, and supplier balances.
Expenses	Records daily shop expenses.
Reports	Supports sales and profit analysis.
Settings & Backup	Maintains shop settings and supports data backup/export.

9. Conclusion

The Maheswari Plywood & Hardware Billing and Inventory Management System combines the shop's important daily activities in one application. The use case model identifies the main users and their responsibilities; the architecture explains how the browser-based application is organized; the ERD presents a logical data model; and

the flowchart and wireframes describe the sales process and major screens. Together, these designs provide a clear reference for understanding, documenting, and further developing the billing system.